

RESEARCH ARTICLE

Strategic Warning in the Cognitive Domain: An Intelligence Framework for Counter-Deception

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ABSTRACT

The literature on countering cognitive warfare is overwhelmingly tactical. This article argues that cognitive warfare achieves strategic effect less at the injection than through the slow re-parameterisation of a society's *zeitgeist*, the epistemic substrate that governs credibility. Reconstituting the problem within Intelligence Studies, as one of strategic warning and counter-deception rather than content moderation, it specifies a target ontology of the epistemic production chain and an institutional-epistemic susceptibility index built from V-Dem. A discrete-time event-history backtest across the world's democracies from 1970 to 2019 shows that the index anticipates subsequent democratic-epistemic regression out of sample. The exercise validates a measurable susceptibility proxy, not the attribution of warfare itself.

KEYWORDS

cognitive warfare; strategic warning; counterintelligence; counter-deception; indications and warning; democratic backsliding; intelligence studies

1 Introduction: the unwatched flank

A defensive doctrine discloses its theory of the threat in what it elects to watch. The body of work now consolidating around the defense of the cognitive domain watches the injection. It watches the deepfake, the coordinated inauthentic network, the engineered outrage, the algorithmic gradient toward the extreme. It assembles these into taxonomies, assigns them identifiers after the fashion of a malware catalogue, and couples each technique to a countermeasure: detect the synthetic artefact, inoculate the citizen against the bias, regulate the recommender.¹ The work is careful, and within its own frame it is useful. It

is also a defense pitched at the wrong altitude, and the consequence of that miscalibration is not a marginal inefficiency but a structural blindness to the decisive operation.

The contention of this article is that cognitive warfare, understood as a strategic instrument rather than a repertoire of tricks, does not seek its decisive effect at the moment any particular falsehood is introduced. At the strategic-warning level, the discrete injection is better treated as noise, unless it is interpreted against the susceptibility of the substrate into which it falls. The decisive effect is the gradual conditioning of that substrate, into which all subsequent injections fall: the prevailing architecture of credibility, salience, and trust by which a population determines what is worth believing and whom it is worth believing. This substrate is what an older vocabulary named the *zeitgeist*. A society whose *zeitgeist* has been conditioned to distrust every institution by reflex, to construe every event as the visible surface of a concealed design, and to sort all information by tribe before it is examined for truth, requires no further deepfakes. It will generate its own and circulate them without external subsidy. The adversary who has reshaped the substrate has carried the campaign before the tactical defender has finished cataloguing the techniques.

This is not an argument that tactical defense is worthless. It is an argument about altitude, and about a corresponding and systematic failure of intelligence. The tactical defender is positioned, funded, and incentivised to perceive the injection. No standing institution is presently charged with perceiving the conditioning of the substrate, because that conditioning is slow, ambient, and, at the granularity of any single event, difficult to distinguish from the ordinary turbulence of public life. It is, in the exact sense established by the founding literature of strategic warning, a signal submerged in noise.² The defense of the cognitive domain, in short, has been miscategorised. It has been treated as a problem of content moderation, platform governance, and media literacy, when it is in the first instance a problem of intelligence, and more precisely a problem of strategic warning and counter-deception. It therefore belongs to Intelligence Studies and to the institutions that practise warning, and it is the central purpose of this article to repatriate it there.

Three considerations make the repatriation substantive rather than terminological. First, warning theory already contains a rigorous account of why slow, structural, strategic threats are systematically under-recognised even when the information sufficient to recognise them is in hand; the cognitive-defense literature has not absorbed that account and continues to rediscover its symptoms one crisis at a time. Second, counterintelligence theory already contains a rigorous account of contests waged through deception, denial, and the deliberate manipulation of an adversary's perception, which is precisely the contest that cognitive warfare instantiates, now conducted against an entire society in place of a single service. Third, the institutions of national intelligence are the only institutions chartered, resourced, and methodologically equipped to sustain surveillance of an adversary's strategic intentions across years rather than across news cycles. To consign

the strategic cognitive flank to content moderators is to leave it unwatched.

The article proceeds as follows. Section 2 establishes the problem of altitude and locates the most sophisticated current frameworks within it, showing that they are pitched below the strategic layer and so leave it undefended. Section 3 reconstitutes the problem within Intelligence Studies, deriving the counter-cognitive mission from the theory of counterintelligence and the classical literature of strategic warning, and engaging the field's own account of what intelligence is and does. Section 4 specifies the target ontology, the epistemic production chain through which a zeitgeist is manufactured and therefore through which it may be subverted and defended. Section 5, the empirical core, operationalises strategic susceptibility from the Varieties of Democracy dataset and subjects the warning claim to a discrete-time event-history backtest across the world's democracies. Section 6 carries the validated instrument forward into the disciplines of stress-testing and the indications-and-warning cycle. Section 7 confronts the governance hazard, because an apparatus competent to guard a society's episteme is, by construction, competent to capture it. The conclusion returns the argument to Intelligence Studies and to the institutional reforms it implies.

The article advances no claim to forecast particular events. Its claim is narrower and, for an intelligence discipline, more defensible: that the strategic susceptibility of a society to cognitive subversion is a measurable, warnable condition; that this condition is at present watched by almost no one; and that the tradition best equipped to watch it is the tradition of strategic warning.

2 The tactical fixation and the problem of altitude

2.1 The tactical frontier and its commitments

The defensive literature is best represented by several serious efforts which, between them, mark the present frontier and, in marking it, expose its boundary.

The first proposes to construct a defense ontology matrix for the cognitive domain by analogy to the frameworks that structure cyber defense.³ The ambition is real, and so is the conceptual borrowing. It enumerates technique classes, visual manipulation, affective triggering, cognitive-bias exploitation, memory manipulation, identity fragmentation, choice-architecture manipulation, assigns each a malware-style identifier, and pairs each with a countermeasure keyed to emotional regulation, bias inoculation, and digital hygiene. A clarification is owed here, because it bears on the user of any such matrix and on a recurrent confusion in the field. The cyber catalogues from which the analogy is drawn are frameworks of *tactics and techniques* validated in a domain, cyberspace, where the adversary's action leaves a forensic trace on a system that can be instrumented. No comparable validated catalogue yet exists for cognition, and the transposition remains aspirational rather than demonstrated; the matrix borrows the *form* of the cyber frameworks without

inheriting their empirical grounding.⁴ The construct inherits the commitments of its model regardless. Its unit of analysis is the technique; its temporal horizon is the engagement; its theory of victory is that if every technique is catalogued and every citizen inoculated, the campaign fails. In its less guarded passages the same proposal drifts toward the idiom of individual persecution, of rogue insiders and the harassment of particular persons, which is not incidental but symptomatic: where the only available unit of analysis is the individual mind under individual attack, the strategic actor and the strategic effect disappear from view altogether.⁵

A second body of work is more rigorous. It models cognitive warfare as a sustained attacker-defender competition over decision advantage, grounded in the Observe-Orient-Decide-Act loop, and it introduces a genuinely important refinement: the distinction between *acute* effects that degrade a decision cycle within minutes to days and *chronic* effects that accumulate over weeks to years and, in the authors' own formulation, "re-parameterise" the acute loops by shaping what is noticed, what is believed credible, and which actions appear legitimate.⁶ This is the nearest the technical literature approaches the argument advanced here, for the chronic re-parameterisation of the substrate is exactly the effect this article places at the centre. Yet the framework's centre of gravity remains the decision loop. Its metrics are decision latency, decision error, and trust calibration; its addressee is the joint-force commander; its theory of victory is decision superiority in a contested environment. The chronic horizon is acknowledged as the condition under which acute contests are fought, but it is not itself constituted as the object to be warned about. The framework, in its own terms, treats the chronic as the weather and the acute as the battle. The wager of this article is that the chronic is the battle, and that it is won or lost years before any acute contest is joined.

Behind both stands the ontological treatment of cognitive competition as a contest across the gradient from peacetime competition through conflict to war.⁷ That work correctly identifies the cognitive domain as a theatre of sustained competition below the threshold of armed conflict, and it supplies the strategic diagnosis. The broader interdisciplinary literature on cyber influence and cognitive threats maps, with care, the psychological and technical mechanisms through which contemporary influence operations are delivered, and the parallel literature on artificial intelligence and military strategy establishes that the automation of content generation and targeting has lowered the marginal cost of the injection toward zero.⁸ What none of this literature supplies is a strategic *defensive* instrument: a means to watch the substrate and to warn before it is lost. The supply of injections having become effectively infinite, a defense premised on interdicting injections has committed itself to an attritional contest it cannot win.

2.2 The altitude model

The shared limitation is not a deficit of rigour. It is a fixation of altitude. Each framework is pitched at the level at which the adversary's action is most visible and least decisive. Figure 1 sets out the resulting structure.

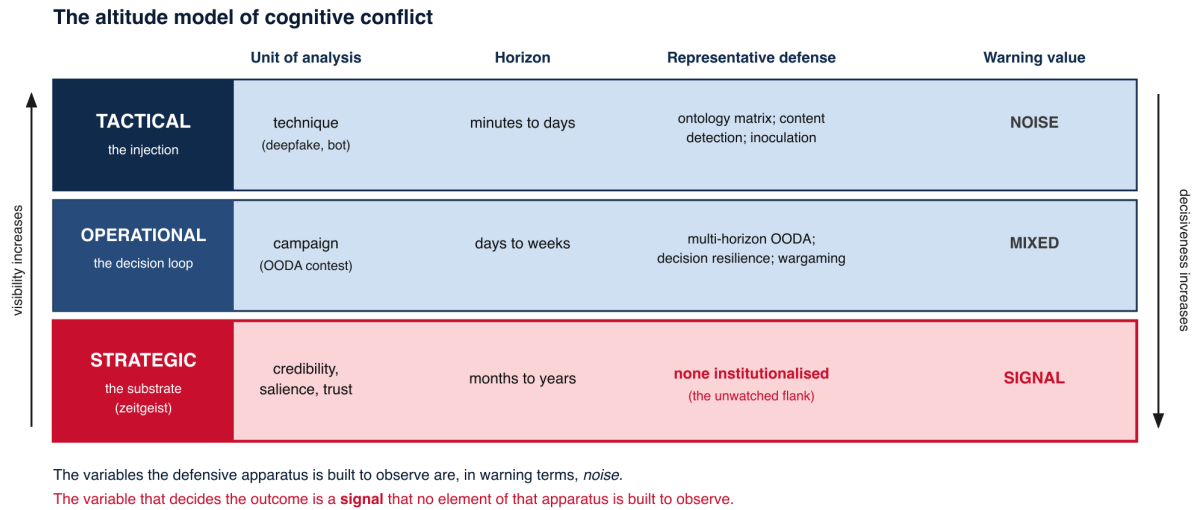


Figure 1. The altitude model of cognitive conflict. Existing defensive frameworks occupy the tactical and operational layers, where the adversary's action is most visible. The strategic layer, where the conditioning of the substrate is decided, has no institutionalised defender, and its indicators, in the vocabulary of warning theory, constitute the genuine signal while the tactical injections constitute noise.

The model renders the omission legible. At the tactical altitude the defender contests the technique and concedes the substrate. At the operational altitude the defender contests the decision loop and inherits the substrate as a given. Only at the strategic altitude does the substrate itself become the object of contest, and at that altitude there is at present no institutionalised defender at all. The enterprise has built two storeys and left the foundation to subside.

The consequence is a specific and recurring pattern of strategic surprise. A society discovers, ordinarily in the aftermath of a contested election, a referendum, a pandemic, or a secessionist mobilisation, that its capacity for shared factual reasoning has degraded to the point where coordinated public action has become difficult. It then searches, retrospectively, for the discrete injections that produced the degradation, and it finds them, because in any open information environment they are always there to be found. It catalogues them, attributes them, sometimes sanctions them, and is surprised again at the next inflection, because it has been watching injections while the substrate subsided beneath them. This is the cognitive-domain expression of the oldest finding in warning theory: that strategic surprise is seldom a failure to collect and almost always a failure

to recognise the significance of what was collected, because the significant signal was structurally indistinguishable from the surrounding noise.⁹

3 Counter-cognitive warfare as a problem in Intelligence Studies

The editorial judgement that the cognitive-warfare literature falls outside Intelligence Studies is, in one respect, well founded. A great deal of that literature is sociology of communication dressed in the vocabulary of conflict. The remedy is not a thicker application of vocabulary. It is to identify the genuine intelligence problem at the core of the cognitive contest and to bring to bear the frameworks the discipline has developed for problems of that structure. Intelligence Studies is not a settled discipline with a single method; it is a field organised, as its principal cartographers have argued, around a small number of recurring projects: the historical reconstruction of secret activity, the definitional and methodological clarification of what intelligence is, the organisational and functional analysis of how it is conducted, and the governance question of how it is to be controlled.¹⁰ The argument that follows is situated deliberately within those projects rather than alongside them.

Two considerations frame the placement. The first is that intelligence has never been confined to the passive collection of secrets. The contemporary field takes as its remit the entire spectrum of secret statecraft, including covert action, black propaganda, and the deliberate shaping of an adversary's perception, and the historical record establishes that the manipulation of belief is among the oldest of intelligence functions rather than a novelty of the digital age.¹¹ The second is that intelligence is, in its classical self-understanding, the production of knowledge vital to national survival, and the reduction of the uncertainty under which the state must decide.¹² If the integrity of a society's collective reasoning is a condition of national survival, and the argument below is that it is, then its surveillance is an intelligence requirement of the first order, however unfamiliar its object.

3.1 Cognitive warfare as a counterintelligence problem

Intelligence Studies has long distinguished the production of knowledge from the protection of the conditions under which knowledge can be produced and acted upon. The latter task is counterintelligence. In its mature theoretical form, counterintelligence rests on a small set of axiomatic propositions: that its purpose is to enable surprise by establishing and maintaining secrecy; that an opposition will employ every available means to collect against the protected entity; and, most pertinently here, that defensive measures alone are never sufficient, so that the contest turns ultimately on an offensive capacity to deceive and to neutralise the opposition.¹³ The function decomposes into four operations: deterrence and detection on the defensive side, deception and neutralisation on the offensive.

The same theory furnishes an unexpectedly exact account of what cognitive warfare *is*.

Prunckun grounds the offensive core of counterintelligence in what he names the “paradox of fiction”: the condition, familiar from literature and cinema, in which a constructed narrative so alters perception that the audience experiences as real what at some level it knows to be fabricated.¹⁴ Cognitive warfare is the industrialisation of the paradox of fiction, scaled from a single deceived service to an entire deceived society. It is, in the strict terminology of the discipline, a strategic deception operation whose target is not an opposing intelligence apparatus but the collective sense-making of a national population. This is the epistemology of secret intelligence turned inside out: the same problem of distinguishing the authentic signal from the planted one, the genuine source from the controlled channel, now confronting not the analyst but the citizen, and at the scale of the whole polity.¹⁵

The recategorisation is consequential because it inverts the defensive question. The content-moderation frame asks how the false content is to be removed. The counterintelligence frame asks how the adversary is to be denied the capacity to run a sustained deception operation against the population, how that operation is to be detected while it runs, and how costs are to be imposed that deter its repetition. These are different questions, with different institutional owners and different success criteria. Table 1 translates the four counterintelligence functions into the cognitive domain and records the current state of coverage.

Table 1. Counterintelligence functions translated to the cognitive domain.

Function	Classical referent	Cognitive-domain analogue	Current coverage
Detection	Catching the spy, the bug, the deception	Identifying an injection or inauthentic network	Strong (tactical)
Deterrence	Raising the cost and lowering the payoff of penetration	Attribution, sanction, and the public hardening of the substrate against payoff	Weak; episodic
Deception	Manipulating the adversary’s perception of the target	Denying the adversary reliable feedback on whether a campaign is shifting the zeitgeist	Effectively absent
Neutralisation	Disabling the opposition’s capability	Degrading foreign production and amplification infrastructure; resilience of the substrate	Partial; contested

The table exposes a second omission beneath the first. Not only is the strategic altitude unguarded; even at the altitudes that are guarded, the enterprise has invested almost exclusively in detection, the single function the counterintelligence tradition regards as

necessary but never sufficient. The offensive functions of deception and neutralisation, which the tradition regards as the real contest, are barely conceptualised for the cognitive domain, in part because they raise the governance hazards examined in Section 7. A defense that consists only of detection has, in counterintelligence terms, conceded the initiative to the attacker in perpetuity.

3.2 Cognitive subversion as a strategic warning problem

If counterintelligence supplies the theory of the contest, strategic warning supplies the theory of the failure. The classical warning literature was constructed to explain a particular and recurring catastrophe: the failure of a forewarned state to recognise an impending strategic attack whose indicators were, in retrospect, abundantly present. Wohlstetter's anatomy of Pearl Harbor established the governing distinction between signals, the indicators that in hindsight pointed to the attack, and noise, the overwhelming volume of competing, ambiguous, and irrelevant information within which the signals were embedded and effectively lost.¹⁶ The warning problem is not fundamentally a collection problem. It is a problem of recognition under an adverse signal-to-noise ratio, compounded by the reluctance of decision-makers to act on warnings that contradict their working assumptions.¹⁷ The discipline's institutional remedy for that reluctance is the deliberate cultivation of counter-opinion: the competitive and alternative analysis, devil's advocacy, and red-team review by which a warning that contradicts the prevailing assumption is kept alive long enough to be tested rather than dismissed.¹⁸

This framing reorders the entire problem. The tactical defender treats the injection as the signal to be detected. Warning theory says the opposite. Injections are abundant, ambiguous, individually inconsequential, and continuously present in any open information environment, which are exactly the properties Wohlstetter assigned to noise. The genuine strategic signal is not any injection but the condition of the substrate that determines whether injections will take: the society's susceptibility. A population whose structural condition has rendered it disposed to cynicism, tribal sorting, and conspiratorial inference is a population in which the next injection will detonate; a population whose substrate is sound is one in which the same injection disperses without effect. The strategic warning question is therefore not what is being injected but whether the substrate is entering a susceptibility window. The remainder of the article is an attempt to make that question answerable, and then to answer it against the historical record.

4 The target ontology: the epistemic production chain

To warn about the substrate one must first specify it as a target: as a structured system with identifiable nodes, flows, and points of penetration, in the manner that a counterintelligence service specifies an adversary's collection apparatus. The substrate is

not an undifferentiated cloud of public opinion. It is produced. A *zeitgeist* is the output of a determinate production chain, and like any production chain it can be mapped, surveilled, and attacked at each stage. This is the target ontology the strategic literature has lacked.

4.1 The chain

The chain has three productive nodes and one emergent product. It runs from the *thinker*, who originates frames, concepts, and interpretive schemata; through the *influencer*, who selects, packages, amplifies, and distributes a subset of those frames; into *society*, the population that adopts, contests, and routinises frames into shared common sense; and it precipitates as the *zeitgeist*, the emergent and relatively durable structure of credibility and salience that then feeds back to constrain what the thinker may think aloud and what the influencer may profitably amplify. Two clarifications guard the ontology against caricature. The influencer node is not confined to the professional content creator; it comprises every actor whose social position amplifies a frame, the teacher, the cleric, the editor, the broadcaster, the local notable, the platform algorithm, so that the function is structural rather than vocational. And society is not a single undifferentiated mass but an array of overlapping clusters, regional, generational, confessional, occupational, that cross-influence one another, so that a frame seeded in one cluster can propagate laterally without ever returning through the formal influencer layer. The state stands outside the chain as its most powerful *operator*: through public policy, official communication, education, and regulation it can act upon every node, to harden the substrate or to capture it, which is the precise reason the governance hazard of Section 7 is constitutive of the problem rather than incidental to it. Figure 2 renders the chain and its penetration points.

Each node has a pedigree in social theory that grounds the ontology and prevents its collapse into journalistic metaphor. The thinker node corresponds to the stratum Gramsci analysed as the producers of a society’s common sense, whose capture is the precondition of hegemony, and the capture of which is now pursued less by persuading thinkers than by discrediting the institutions, above all the universities, that certify them.¹⁹ The influencer node corresponds to the gatekeeping and amplification function Lippmann identified as the manufacture of consent through the management of the pictures in our heads, and that Bernays converted into a profession.²⁰ The society node, and specifically the dynamics by which dissenting frames are silenced or emboldened, corresponds to Noelle-Neumann’s spiral of silence, the mechanism by which perceived majority opinion suppresses the public expression of minority views and so accelerates its own apparent dominance.²¹ The semiotic literature on strategic conspiracy narratives supplies the contemporary mechanism by which all three nodes are attacked at once, through the propagation of a master interpretive frame that recodes every institution as a deceiving adversary and so

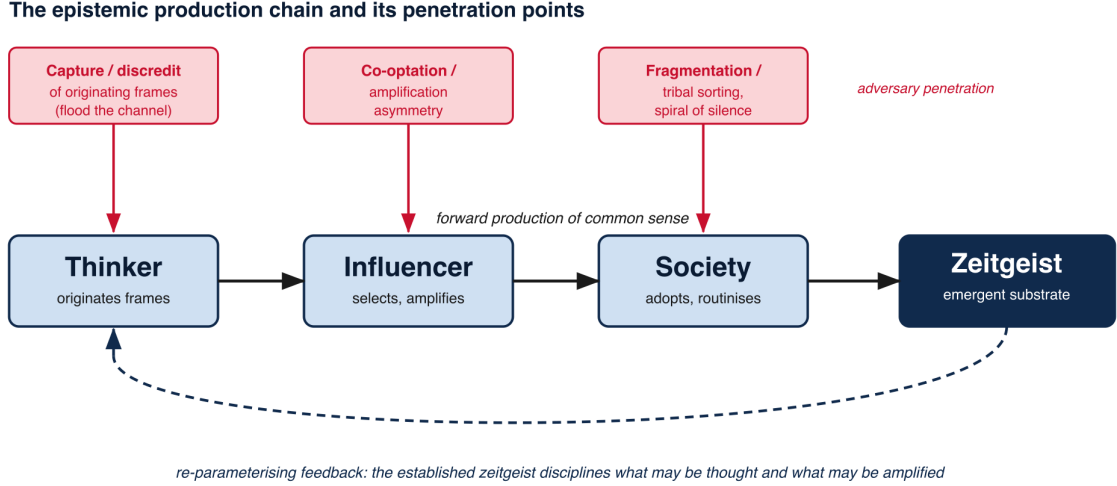


Figure 2. The epistemic production chain and its adversary penetration points. Forward arrows trace the production of common sense; the dashed arrow traces the re-parameterising feedback by which an established zeitgeist disciplines the upstream nodes.

seals the substrate against correction.²²

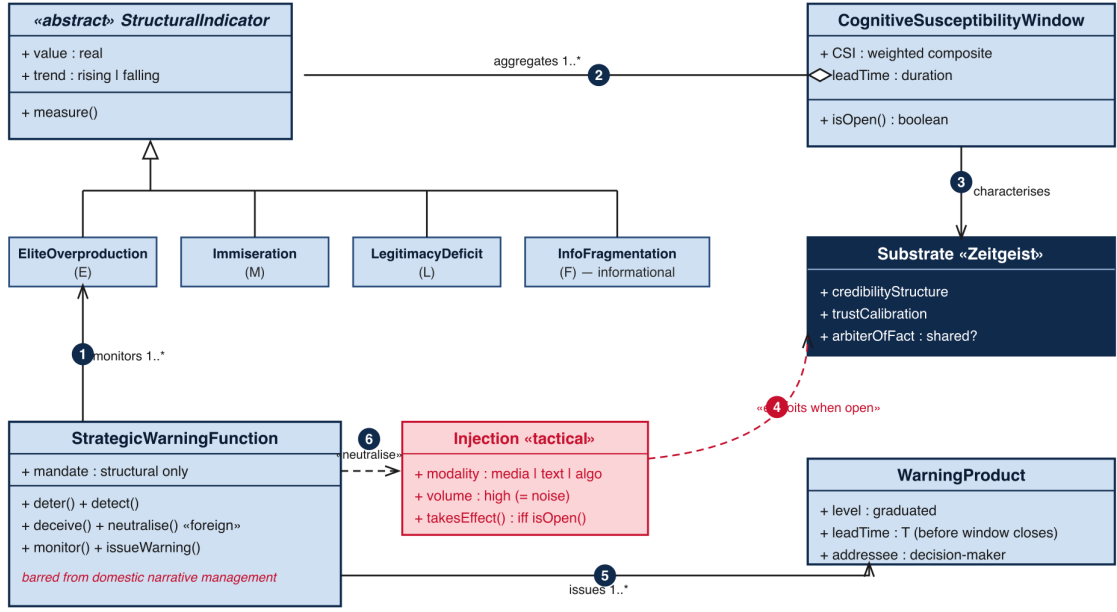
The relationship between this ontology and the warning instrument is best made precise. Figure 3 expresses the strategic-warning ontology as a class model: the structural indicators aggregate into a susceptibility window that characterises the substrate; the tactical injection takes effect only when that window is open; and the warning function monitors the indicators and issues warning, while remaining, by deliberate construction, barred from the domestic narrative management that would convert defense into offense against one’s own population.

4.2 Penetration points and the threat-mitigation matrix

The strategic value of specifying the chain is that it converts a diffuse anxiety about disinformation into a discrete set of penetration points, each with its own threat mechanism, strategic mitigation, warning indicator, and counterintelligence function. Table 2 sets out the resulting matrix, which is what an indications-and-warning cell tasked with the cognitive substrate would actually maintain.

Two features of the matrix are decisive. First, the mitigations at every node are structural and institutional, not informational. They concern the architecture of credibility, not the truth-value of particular messages. This is what it means to defend the substrate rather than the injection. Second, the warning indicators in the fourth column are, without exception, slow-moving structural quantities rather than fast event counts. They are the kind of variable no content-moderation dashboard tracks and no news cycle surfaces, which is precisely why the strategic flank is unwatched. To watch it requires an instrument built for structural, slow-onset warning, and the burden of the remainder of the article is

Strategic cognitive-warning ontology (UML)



Sequence: (1) monitor indicators; (2) aggregate to susceptibility window; (3) characterise substrate; (4) injection exploits open window [adversary]; (5) issue warning; (6) neutralise foreign infrast

Figure 3. The strategic cognitive-warning ontology expressed as a UML class model. In contrast to a tactical technique catalogue, the model places the susceptibility window, not the injection, at the centre, and constitutes the injection as effective only conditional on the window being open.

to construct such an instrument and to demonstrate, against the historical record, that it carries genuine predictive content.

4.3 From the chain to a measurable susceptibility

A precise vocabulary is required before the construct can be measured, because the field routinely conflates five distinct things, and the conflation is the source of most overclaiming about cognitive warfare. *Vulnerability* is the standing, slowly changing disposition of a society to epistemic harm, the analogue of terrain. *Susceptibility* is the time-varying elevation of that disposition, the window during which the terrain is briefly traversable; it is what the index below measures. *Exposure* is the volume and reach of hostile material actually directed at the society. *Adversary action* is the deliberate agency of a hostile planner who times exposure to coincide with an open susceptibility window. And *outcome* is the realised degradation of the substrate, observable only after the fact. Cognitive warfare, in the strict sense, is the conjunction of adversary action and exposure against an open window. The instrument constructed in this article measures one term in that sequence, susceptibility; it does not measure exposure or adversary action, and it cannot by itself separate a substrate degraded by a hostile campaign from one degraded by endogenous decay. This is a deliberate and consequential boundary, returned to in Section 5.7, and it is the reason the empirical claim below is confined to susceptibility and its

Table 2. Threat vectors and strategic mitigations by chain node.

Node	Penetration mechanism	Strategic mitigation	Warning indicator	Function
Thinker	Capture or discredit the originators of frames; flood the channel so authority cannot be distinguished from noise	Protect the institutions of certified knowledge; sustain independent, trusted sense-making	Collapse in measured institutional trust; undifferentiated cynicism	Deter / Neutralise
Influencer	Engineer amplification asymmetries; counterfeit or co-opt high-reach intermediaries; reward polarising frames structurally	Diversify and harden amplification; remove the structural reward for polarisation	Concentration of reach; affective polarisation of dominant frames	Detect / Neutralise
Society	Accelerate tribal sorting; trigger spiral-of-silence cascades; normalise conspiratorial inference as the default	Maintain cross-cutting affiliations; protect the expressibility of dissent within shared norms	Sorting indices; share of population reading events conspiratorially	Detect / Deter
Zeitgeist	Lock in a self-sealing frame that recodes all institutions as deceptive, foreclosing correction	Preserve a shared adjudicative substrate: agreed procedures for settling fact	Loss of any commonly accepted arbiter of fact; symmetric distrust of all sources	Deception / Neutralise

association with subsequent regression, and stops short of the attribution of warfare.

The decisive theoretical claim is that a society’s susceptibility to cognitive subversion is not principally a function of how many falsehoods are injected into it but of its structural condition at the time of injection. The same campaign that fails against a cohesive society succeeds against a stressed one. Susceptibility is therefore a property of the host, not of the pathogen, and it is a property that varies slowly, structurally, and measurably. The most developed account of the structural conditions of societal instability is the structural-demographic theory associated with cliodynamics, which holds that the propensity of a complex society toward internal breakdown rises and falls with a small number of drivers, principally the immiseration of the broad population relative to expectations, the overproduction of aspirant elites competing for a fixed number of positions of status and power, and the consequent erosion of the legitimacy and fiscal health of the state.²³ The empirical and predictive standing of that theory is disputed, and nothing in what follows depends on accepting it as a law of history. What it supplies, and what the

cognitive-defense literature wholly lacks, is a principled and measurable account of why a society’s substrate becomes susceptible, independent of any particular adversary action.

The bridge to warning theory is now exact. Structural stress is the slow, ambient, easily dismissed signal. The flood of daily injections is the noise. Let the cognitive susceptibility of a society be understood as an increasing function of three structural quantities and one informational quantity:

$$\text{CSI} = w_1E + w_2M + w_3L + w_4F, \quad (1)$$

where E indexes elite overproduction and intra-elite competition, M popular immiseration relative to expectations, L the legitimacy deficit of governing institutions, and F the fragmentation of the information environment, the degree to which the society lacks any commonly accepted arbiter of fact. The weights w_i are not universal constants to be discovered but parameters to be calibrated against the historical record of each society, which is the function of the backtesting discipline below. Equation (1) is deliberately the only formalism in this article. Its purpose is not to compute a society’s fate but to discipline the warning watch list: to specify what must be measured, to name the four quantities whose joint elevation defines a susceptibility window, and to insist that the fourth, informational fragmentation, is the variable through which the adversary acts while the first three determine whether that action takes. The next section subjects the informational and institutional components of this construct, the components a cross-national dataset can actually measure, to empirical test.

5 Empirical validation: a V-Dem backtest of strategic susceptibility

For a warning framework to contribute beyond conceptual reframing, it must be exposed to historical backtesting. The discipline developed in the quantitative risk professions for validating warning models, backtesting against the historical record and stress-testing against adversarial scenarios, transfers to this problem and supplies the falsifiability that warning theory has historically lacked. This section conducts the backtest. The question is precise and prospective: does a measure of a society’s institutional-epistemic susceptibility, computed from data available at a given time, anticipate a subsequent democratic-epistemic regression before it occurs? If it does, the warning claim has content; if it does not, the framework is theory without traction, and it is better to know. Two boundaries should be fixed at the outset, because they govern what the result can and cannot mean. First, of the four drivers of the susceptibility construct, V-Dem permits a test of only two, the legitimacy deficit L and the informational fragmentation F ; the structural-demographic drivers E and M lie outside the dataset and are not tested here. Second, and following the vocabulary fixed in Section 4.3, the backtest measures

susceptibility, not exposure or adversary action; it therefore validates a measurable proxy associated with subsequent regression, and does not, and cannot, by itself attribute that regression to cognitive warfare. The honest object of the exercise is the warnable condition, not the warfare.

5.1 Data and scope

The analysis uses the Varieties of Democracy (V-Dem) dataset, version 16, which provides expert-coded annual measures of the institutional, informational, and accountability dimensions of political order for some two hundred polities from 1789 to 2025.²⁴ The index components are constructed from the V-Dem Core indices; economic and regime-classification covariates are drawn from the Country-Year Full+Others release. The analysis window is 1970 to 2019, the lower bound chosen for the density and reliability of post-1970 coding and the upper bound chosen so that the five-year-ahead outcome remains observable through 2024. The unit of analysis is the country-year. The use of a genuinely global dataset is deliberate and answers a standing parochialism in the field, which has tended to treat cognitive and informational subversion as an affliction of the Western democracies; the structural susceptibility theorised here is not regionally bounded, and the analysis includes the post-conflict and transitional states in which the politicisation of intelligence and the contest over the informational order are frequently most acute.²⁵

5.2 Operationalizing institutional-epistemic susceptibility

The full construct of Equation (1) cannot be measured from V-Dem alone, because two of its drivers, elite overproduction (E) and popular immiseration (M), are structural-demographic quantities that require economic and demographic data outside the dataset's scope. What V-Dem measures with precision is the institutional-epistemic projection of the construct: the legitimacy and accountability of institutions (L) and the integrity of the informational environment (F). The empirical index is therefore named, with deliberate exactness, the Institutional-Epistemic Susceptibility Index (IESI), and is understood as the measurable projection of the broader susceptibility construct rather than its whole. The deferred drivers E and M are returned to in Section 5.7.

The IESI is built from six V-Dem Core indices, each scaled from zero to one, each oriented so that higher values denote a healthier substrate, and combined so that higher IESI denotes greater susceptibility. The informational component draws on the index of alternative sources of information and the freedom-of-expression and alternative-information index, which jointly capture the presence or absence of a plural, uncensored informational environment, the operational meaning of a shared arbiter of fact. The institutional component draws on the rule-of-law index, legislative and judicial constraints on the executive, and the regime-corruption index, which jointly capture the horizontal accountability whose erosion constitutes the legitimacy deficit. The index is the complement of the mean of

these components, so that

$$\text{IESI} = 1 - \overline{(\text{informational integrity, institutional accountability})},$$

computed for each country-year with at least five of the six components present. The construction is deliberately austere: the six components are V-Dem indices already expressed on a common zero-to-one scale, so they are equally weighted and averaged without prior re-standardisation, and only the resulting index is standardised to unit variance when it enters the model as a predictor; the five-of-six rule retains country-years with a single missing component while excluding those too sparse to characterise. The exact V-Dem variable names, scales, coding directions, and missingness, together with the full operational definition and a replication note, are reported in the methods appendix. Because two of the informational components are also constituents of the electoral-democracy index used to define the outcome below, the analysis additionally constructs an accountability-only variant of the IESI from the four institutional components alone, none of which enters the outcome measure. This variant is not a footnote but a central test: if the accountability-only index predicts the outcome, the result cannot be an artefact of definitional overlap between predictor and outcome.

5.3 Defining cognitive-political inflection

The dependent event is the onset of a cognitive-political inflection, operationalised as the beginning of a substantial and sustained regression in the electoral-democratic order. A country-year is coded as an inflection onset if the electoral-democracy index declines by at least 0.10, on its zero-to-one scale, over the subsequent five years, and if the preceding year was not already within such a decline; the onset is thereby dated to the first year of the episode, and years internal to an ongoing decline are removed from the risk set so that a single episode contributes a single onset.²⁶ This is the political surface of the cognitive contest: a measurable break in the order through which a society governs itself by shared procedures. The theoretical claim is that the epistemic substrate degrades first, that this degradation is the warning signal, and that the political inflection breaks later, the lag between them being precisely the warning lead time the instrument is designed to recover.

The analysis population is restricted to the world’s democracies, defined by the V-Dem Regimes of the World classification as electoral and liberal democracies, a principled and widely used cutpoint.²⁷ The restriction is dictated by the structure of the event rather than by the search for a result: a society can suffer the erosion of a democratic-epistemic order only if it possesses one, and the closed and electoral autocracies that sit at the floor of the scale cannot register the decline. The boundary condition this imposes, that the instrument speaks to the susceptibility of functioning democracies and not to the dynamics of consolidated autocracies, is returned to in Section 5.7.

5.4 Backtesting design

The estimator is a discrete-time event-history (survival) logit, the standard tool for modelling the onset of a binary event in panel data. Each democracy contributes one observation per year that it remains at risk, that is, every year in which it is a democracy and not already inside a decline episode; the binary outcome records whether an inflection onset begins in that year; and the model estimates the conditional probability of onset as a logistic function of the susceptibility index measured in the prior year, together with controls. The principal predictor is the one-year lag of the IESI, standardised to unit variance so that its coefficient expresses the change in onset odds associated with a one-standard-deviation increase in susceptibility. The controls are the log of GDP per capita, lagged, to absorb the confound of economic development, and a quadratic in calendar time to absorb period effects, above all the global wave of democratic regression after 2010. Standard errors are clustered by country throughout, because the repeated observations of a single polity are not independent. The main specification deliberately omits any control for the contemporaneous level of democracy, which is collinear with the susceptibility index by construction and would partial out the very variation the index is meant to capture; a conservative specification that includes it is reported alongside, and its attenuating effect is interpreted rather than concealed.

The decisive test of a warning instrument, however, is not its fit to the data on which it was estimated, which is retrospective, but its performance on data it has not seen, which is prospective and is what warning actually requires. The analysis therefore estimates the model on country-years through 2009 and uses the resulting coefficients to predict inflection onsets over 2010 to 2019, reporting the out-of-sample discrimination as the headline result. This is a demanding test, for the prediction window is precisely the autocratization wave, the period in which the global pattern of democratic regression was at its most intense and its least anticipated. Figure 5 sets out the resulting indications-and-warning cycle, of which the backtest is the validation stage.

5.5 Results

Among the world’s democracies from 1970 to 2019, the panel comprises 3,174 country-years at risk, 115 countries, and 60 inflection onsets, a base rate of 1.9 per cent, which is to say that the event is rare and the prediction problem correspondingly hard. Table 3 reports the estimates.

The result is clear and, given the stakes of the argument, worth stating without hedging where hedging is not warranted and with hedging where it is. In the main specification, a one-standard-deviation increase in institutional-epistemic susceptibility is associated with an increase in the odds of a subsequent inflection onset by a factor of 1.80, an effect significant at the one-in-a-thousand level, and the model discriminates onsets from non-onsets with an area under the receiver-operating curve of 0.82. The

Table 3. Discrete-time event-history logit of cognitive-political inflection onset, democracies, 1970–2019. Cells report log-odds coefficients with country-clustered standard errors in parentheses; the susceptibility index is standardised. Odds ratios for the susceptibility coefficient are given in the text.

	(1) Full IESI	(2) Accountability- only	(3) Conservative
Susceptibility index (IESI), $t - 1$, z	0.586*** (0.166)	0.558** (0.183)	0.269 (0.193)
Electoral democracy level, $t - 1$			−3.520* (1.486)
log GDP per capita, $t - 1$	−0.319 (0.201)	−0.318 (0.208)	−0.198 (0.200)
Calendar time (linear, centred)	0.467** (0.150)	0.449** (0.150)	0.525*** (0.147)
Calendar time (quadratic)	0.160* (0.074)	0.169* (0.074)	0.138 (0.077)
Country-years	3,174	3,174	3,174
Onsets	60	60	60
Countries	115	115	115
In-sample AUC	0.816	0.813	0.816

Note: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. Standard errors clustered by country.

accountability-only index, which by construction shares no component with the outcome and so cannot be charged with circularity, yields an almost identical estimate, an odds factor of 1.75, significant at the one-in-five-hundred level, with discrimination of 0.81. This is the result that matters most: the predictive content of susceptibility survives the removal of every component that overlaps with the outcome, which establishes that the finding is not an artefact of measurement. The conservative specification, which adds the contemporaneous level of democracy, attenuates the susceptibility coefficient to an odds factor of 1.31 and to conventional non-significance, exactly as collinearity between susceptibility and the level of democracy would predict; the attenuation is a property of the over-controlled specification and not evidence against the relationship, and it is reported here so that the reader may judge rather than be managed.

The out-of-sample test is the more severe and the more important. A model estimated only on country-years through 2009, and then asked to predict the inflection onsets of the 2010s wholly unseen, discriminates the onsets of 2010 to 2019 with an area under the curve of 0.70 for the full index and 0.70 for the accountability-only index. A warning

instrument that, trained only on the past, ranks the future democracies that will suffer an inflection above those that will not, seven times in ten, is not a forecasting oracle and does not pretend to be; it is, however, exactly the kind of graduated structural warning that the indications-and-warning tradition has always sought and seldom possessed. The discrimination is a ranking property, and it should not be confused with calibrated probability: as Section 5.6 reports, the instrument orders democracies well but its absolute-probability accuracy barely improves on the base rate, which is the expected and honest signature of a rare-event warning model. Figure 4 renders the underlying dynamics directly, plotting the susceptibility trajectories of a set of illustrative cases with their inflection onsets marked.

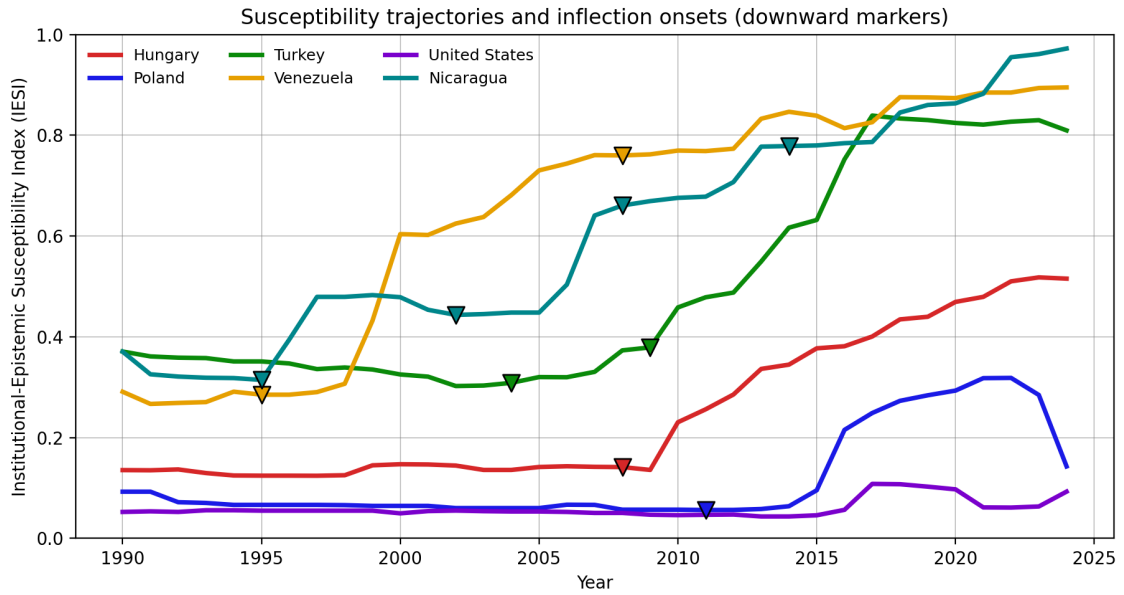


Figure 4. Institutional-epistemic susceptibility trajectories, 1990–2024, for six illustrative democracies, with inflection onsets marked by downward triangles. In the major modern episodes of regression the susceptibility index was already elevated relative to consolidated democracies before the onset; in the EU cases the erosion proceeded from a lower base and the index rose steeply through the episode; the United States exhibits a rising trajectory without a formal inflection at the stated threshold.

5.6 False positives, false negatives, and warning value

A warning instrument is used by setting a threshold and acting on the warnings it issues, and its value is therefore properly assessed not by a single fit statistic but by the structure of its errors at an operating point. Translating the out-of-sample predictions into a binary warning by flagging the country-years in the highest quartile of predicted risk yields the confusion structure in Table 4.

The structure repays interpretation, because it expresses with unusual clarity the characteristic dilemma of strategic warning. Flagging the highest-risk quartile of democracies catches half of all inflection onsets one to five years before they begin, at the cost of a

Table 4. Warning performance, out-of-sample (2010–2019), at a top-quartile risk threshold.

	Inflection onset	No onset
Warning issued (top-quartile risk)	17 (true positive)	197 (false positive)
No warning	17 (false negative)	622 (true negative)
Hit rate (recall) = 0.50 False-alarm rate = 0.24 Precision = 0.08		

false-alarm rate of roughly one in four. The precision is low, eight per cent, and the low precision is in part a direct consequence of the rarity of the event: when the base rate is around four per cent in the test window, even a strong predictor will generate many false positives for each true one, and any warning system calibrated to catch rare catastrophes must accept some version of this arithmetic. But the figure should not be waved away. One hundred and ninety-seven false positives is not a rounding detail; for an intelligence or governance consumer it means that the large majority of democracies flagged in any given window will not in fact cross the threshold, and an instrument used as if a warning were a prediction would squander credibility and invite exactly the politicised misuse Section 7 anticipates. The disciplined reading is therefore narrow: the warning is a trigger for analytic attention and institutional hardening, never a warrant for action against a flagged society, and its value rests not on precision but on the lift it provides over the base rate, which is real but bounded: a warned democracy is several times more likely to suffer an inflection than an unwarned one. This is the same trade-off that has always governed strategic warning, from missile attack to financial crisis, and the recognition that a warning need not be a forecast to be valuable is among the oldest lessons of the discipline. A false positive is not a failure of the instrument but a demand for analytic attention; a false negative, the inflection that arrives unwarned, is the failure the instrument exists to reduce, and it reduces it by half.

Discrimination alone overstates what a single threshold conveys, and a reviewer is entitled to the harder diagnostics. Three are decisive, and none is flattering by accident. The out-of-sample precision-recall area is 0.091 against a no-skill baseline of 0.040, so the instrument carries roughly two-and-a-third times the precision a random ranker would, which is genuine but modest lift on a rare event. The out-of-sample Brier score is 0.0382, essentially indistinguishable from the climatological Brier of 0.0383 obtained by predicting the base rate for every case: the model orders democracies far better than it estimates their absolute probabilities, and its calibrated risk numbers should not be reported to two decimal places as if they were forecasts. What the model does well is ordering, and on ordering it is well behaved. Table 5 bins the out-of-sample country-years into quintiles of predicted risk and reports the realised onset rate in each: the rate rises monotonically from well below the base rate in the safest quintile to roughly double it in the most-at-risk, which is the behaviour a usable warning ranking must exhibit. In

cost-loss terms the instrument is therefore suited to a consumer whose cost of a missed inflection greatly exceeds the cost of an unnecessary analytic look, and unsuited to any consumer contemplating consequential action on the strength of a flag.

Table 5. Out-of-sample calibration (2010–2019, full IESI), by quintile of predicted risk.

Predicted-risk quintile	Q1 (safest)	Q2	Q3	Q4	Q5 (riskiest)
Mean predicted onset probability	0.005	0.008	0.017	0.038	0.106
Observed onset rate	0.006	0.018	0.041	0.059	0.076

Note: 853 country-years; test-window base rate 0.040. PR-AUC = 0.091; Brier = 0.038 (climatology 0.038).

A reviewer will also ask whether the index merely restates democratic weakness through its informational components. Two further results, reported in full in the methods appendix, bound that concern. A predictive horse-race that enters the informational and institutional sub-indices separately assigns the clean predictive weight to the institutional, accountability dimension, not the informational one, which is reassuring precisely because the informational components are the ones that overlap with the outcome. And a leave-one-component-out exercise shows the odds factor moving only between 1.70 and 1.92 as each of the six components is removed in turn, so no single variable, and in particular no single informational variable, is driving the result.

The result is moreover robust to the definition of the event. Table 6 reports the estimated odds factor for the susceptibility index across a grid of onset definitions, varying the decline threshold and the window over which it is measured. The coefficient is positive and significant in every cell, ranging narrowly from 1.70 to 1.88, which establishes that the finding is not an artefact of a single knife-edge operationalisation of the outcome.

Table 6. Robustness of the susceptibility effect to the onset definition. Cells report the odds factor for a one-standard-deviation increase in IESI; all coefficients significant at $p < 0.001$.

Decline threshold	4-year window	5-year window	6-year window
0.08	1.82	1.71	1.71
0.10	1.83	1.80	1.83
0.12	1.88	1.88	1.81

5.7 Limits of the V-Dem test

Intellectual honesty requires that the limits of the test be stated as plainly as its results, and they are more confining than the headline numbers suggest. Six limits are material. First, the IESI measures only the institutional-epistemic projection of the susceptibility construct; the two structural-demographic drivers of Equation (1), elite overproduction

and popular immiseration, lie outside V-Dem and are not tested here, so the validation is a validation of half the model, and a fuller test awaits the integration of economic and demographic series. Second, and most consequentially, the test measures susceptibility and not warfare. It includes no variable for exposure or adversary action, no foreign-influence operation, platform-manipulation, or media-capture measure, and it therefore cannot distinguish a substrate degraded by a hostile campaign from one degraded by purely endogenous decay. What the backtest validates is a vulnerability, in the strict sense of Section 4.3, not the attribution of an attack; the inferential distance between the two is the principal boundary of the entire empirical exercise. Closing that distance would require external series on exposure and adversary activity, foreign-influence operations, platform manipulation, media capture, polarisation and trust surveys, joined to the susceptibility measure; that integration is the necessary route to any claim pitched at the level of warfare rather than vulnerability, and it is deliberately left to future work rather than asserted here. Third, the instrument speaks to the susceptibility of democracies and is silent, by the logic of the room-to-fall problem, on the dynamics of autocracies, where the cognitive contest takes a different and equally consequential form; the framework is general but the present test is not. Fourth, the outcome is a political surface, the regression of the electoral-democratic order, used as an observable proxy for the underlying epistemic inflection; the two are theorised to be coupled, but they are not identical, and the outcome is more accurately described as democratic-epistemic regression than as a direct measure of substrate degradation. Fifth, the theory’s central temporal claim, that the informational substrate degrades *first* and the institutional order *later*, is not corroborated by the data. A cross-lagged panel test finds no significant lead of informational over institutional susceptibility in either direction, and the predictive weight rests on the institutional dimension; the substrate-first ordering therefore remains a theoretical conjecture, not an established finding, and the language of the article is calibrated accordingly. Sixth, expert-coded data carry the possibility that the same global events which drive regression also shift expert perceptions of the antecedent indicators, a form of contamination that the out-of-sample and accountability-only tests mitigate but cannot wholly exclude. None of these limits is individually fatal, but together they bound the claim tightly: a structural measure of institutional-epistemic susceptibility, computed from data available in advance, carries genuine and out-of-sample predictive content for subsequent democratic-epistemic regression. That is less than a validation of cognitive warfare and more than a conceptual reframing, and it is the most the present evidence will bear. The order breaks later than the susceptibility is measurable, and the interval between them is warning time that is at present going unused.

6 From validation to warning practice

The backtest establishes that the signal exists. It does not by itself constitute a warning capability, which requires that the validated instrument be embedded in a standing process and complemented by a forward-looking discipline that the historical record cannot supply. Figure 5 sets out the indications-and-warning cycle in which the instrument would operate.

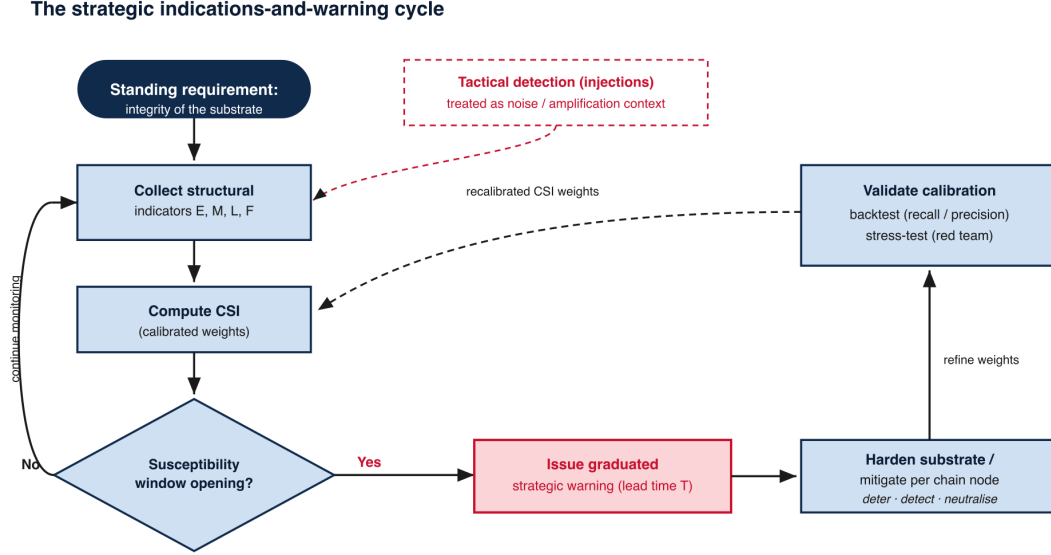


Figure 5. The strategic indications-and-warning cycle. Structural indicators are collected and composited into a susceptibility index; a threshold crossing issues a graduated warning with a stated lead time; mitigation is directed at the chain node under greatest stress; and the instrument is continuously recalibrated by backtesting and stress-testing. Tactical detection of injections enters not as the primary signal but as context.

6.1 Indicators, signal, and noise

The cycle begins from the watch list, and the watch list is what most sharply distinguishes the strategic instrument from the tactical apparatus it relativises. Table 7 contrasts the two indicator ontologies and classifies each by its Wohlstetter type.

The table compresses the article’s thesis. The variables the entire defensive apparatus is built to observe are, in warning terms, noise; the variable that determines the outcome is a signal that no element of that apparatus is built to observe, and that the preceding section has shown to be measurable and predictive. Correcting this is not a matter of better detection. It is a matter of standing up a function, at the strategic altitude, whose requirement is the integrity of the substrate.

Table 7. Indicator ontology: tactical noise versus strategic signal.

Layer	Indicator class	Observable by	Lead time	Type
Tactical	Synthetic-media volume; inauthentic-account activity; viral falsehood counts	Platforms; fact-checkers	Hours to days	Noise
Operational	Narrative-campaign coherence; cross-platform coordination; framing convergence	Open-source analysts	Days to weeks	Mixed
Strategic	Institutional-epistemic susceptibility (IESI) and its structural-demographic complements (E , M)	A national warning function	Months to years	Signal

6.2 Stress-testing against the adaptive adversary

Backtesting looks backward, against the historical record; stress-testing looks forward, against an intelligent adversary, and supplies what the historical record cannot. The instrument is subjected to red-team scenarios in which a hostile planner, given the same structural map of the target society, designs the campaign most likely to convert a given level of structural stress into durable substrate damage. Stress-testing serves two functions. It discloses the non-linearities and tipping points that a purely historical calibration will miss, because the adversary is actively seeking the threshold the defender hopes is distant. And it converts the warning instrument into a planning instrument: by revealing which structural variable the adversary would most want to move, it identifies which variable the defender must most urgently protect. Should the red team consistently find that informational fragmentation is the cheapest lever by which to convert latent structural stress into manifest substrate collapse, the defense of the commonly accepted arbiter of fact would become the priority mitigation, ahead of any particular content intervention. This is the natural conjecture of the framework, but it must be entered as a conjecture and not as a finding: the backtest of Section 5 does not isolate F as the operative driver, and if anything assigns the cleaner predictive weight to the institutional accountability dimension, so the proposition that fragmentation is the decisive lever is a hypothesis for the stress-test to examine rather than a result it may assume. This is the point at which the strategic instrument rejoins, and reorders, the tactical enterprise from which the argument began. The design of such a function, however, is constrained by an institutional reality the warning literature knows well: any standing capability must reconcile incompatible imperatives, the bureaucratic demand for stable procedure against the operational demand for flexibility, and it will function only to the degree that it is built for pragmatic adaptation rather than for doctrinal purity.²⁸

7 The governance hazard: who guards the zeitgeist?

An apparatus competent to monitor the structural condition of a society's collective reasoning, and to act to protect it, is by construction an apparatus competent to manage that reasoning to the advantage of whoever controls it. This is not a peripheral caveat. It is the central reason the strategic flank has remained unwatched, and any proposal to watch it that does not confront the hazard directly is not serious.

The hazard has a precise name in Intelligence Studies: the politicisation of intelligence, the corruption of the function that occurs when the production of assessments is bent to serve the policy preferences or the power of the principal rather than the truth, with the consequence that intelligence becomes either irrelevant or captured.²⁹ A cognitive-warning function is singularly exposed to it, for two reasons. First, its object, the integrity of the substrate of public reasoning, is inseparable from the content of public reasoning, so that the line between protecting the conditions of debate and policing the debate itself is thin and, under pressure, vanishes. Second, the incumbent power is always the actor with the most to gain from defining the prevailing zeitgeist as healthy and every challenge to it as subversion. A cognitive-warning apparatus captured by an incumbent is not a defense against cognitive warfare. It is cognitive warfare, conducted by the state against its own population under the colour of defense, and the historical record of the security state's management of public knowledge gives ample warrant for the fear.

That record is not hypothetical, and the Intelligence Studies literature has documented it in detail. The modern state's long obsession with secrecy, and its discovery that information it could no longer suppress could instead be managed, produced over the twentieth century a sophisticated practice of offensive information management, in which governments unable to keep their secrets increasingly sought to offset damaging disclosure with their own micro-managed publications and narratives.³⁰ The same literature documents the countervailing force: the memoirs, the leaks, the whistleblowers, and the public reckonings through which the intelligence state's management of the record has been contested and, intermittently, checked.³¹ The lesson of that history is not that the state should never concern itself with the condition of public knowledge, which it has always done and will always do, but that such concern is perennially liable to slide from the protection of the public's capacity to reason into the management of what the public is permitted to conclude. The accountability and ethics of secret activity are, for just this reason, not an addendum to the study of intelligence but one of its constitutive projects.³²

The argument of this article does not dissolve the hazard, because it cannot be dissolved. It can only be governed, and three principles, drawn directly from the established theory and practice of intelligence oversight, are necessary conditions of any legitimate cognitive-warning function. The first is a structural rather than a substantive mandate: the function must be chartered to monitor and protect the structural indicators of substrate integrity,

the institutional and informational quantities of the validated index and their structural-demographic complements, and explicitly forbidden from adjudicating the truth-value or acceptability of particular frames, narratives, or speakers. It watches the rule of law, the plurality of information, the accountability of institutions; it does not watch opinions. The second is adversarial transparency: the indicator set, the thresholds, and the warnings issued must be subject to the same independent legislative and judicial oversight, with the same access, that governs the rest of the intelligence enterprise, precisely because the temptation to capture is greatest here. The third is an asymmetry of permitted action: the function may warn, may harden institutions, and may, under strict authorisation, neutralise demonstrably foreign infrastructure, but it must be barred by construction from the domestic deception and narrative-management operations that the counterintelligence repertoire would otherwise license. The offensive functions legitimate against a foreign service are illegitimate against one's own population, and the apparatus must be built so that it cannot turn inward even when its controllers wish it to.

These constraints will strike the operator as crippling and the civil libertarian as insufficient, which is the characteristic signature of a genuine constitutional settlement rather than a slogan. The point is not that the hazard is small. The point is that the alternative, leaving the strategic substrate undefended on the ground that defending it is dangerous, is itself a decision with consequences, and an adversary that has grasped the altitude argument of this article will exploit precisely that abdication. To decline to build the warning function is not to remain neutral. It is to leave the most consequential terrain of contemporary conflict to whichever actor is least inhibited about occupying it.

8 Conclusion

The literature on countering cognitive warfare has built sophisticated instruments for watching the wrong altitude. It watches the injection, the technique, the discrete falsehood, because these are visible, catalogable, and concrete. It does not watch the substrate, the structural condition of a society's collective reasoning, because that condition is slow, ambient, and indistinguishable at any moment from the ordinary noise of public life. Yet the substrate is where cognitive warfare achieves its strategic effect, and a defense that does not watch it is a defense of an unguarded flank.

Correcting the error is not a matter of better detection but of recategorisation. Counter-cognitive warfare is not a content-moderation problem. It is an intelligence problem, and specifically a problem of strategic warning and counter-deception, for which Intelligence Studies already possesses the necessary apparatus. The theory of counterintelligence supplies the structure of the contest, an industrialised deception operation against a national population, fought through the functions of deterrence, detection, deception, and neutralisation. The classical theory of strategic warning supplies the structure of the

failure, the loss of the decisive signal, the susceptibility of the substrate, within the noise of the daily injection. The target ontology of the epistemic production chain converts the diffuse substrate into a surveillable system of nodes and penetration points. And the empirical core of the article supplies something the strategic literature has rarely attempted: a backtest, against the historical record of the world's democracies and out of sample, showing that a structural measure of institutional-epistemic susceptibility anticipates subsequent democratic-epistemic regression. The claim is bounded, and the article has been at pains to bound it. The exercise validates a measurable susceptibility, not the attribution of an attack; it tests two of the construct's four drivers, not all; and the conjecture that the informational substrate degrades before the institutional order is one the data do not yet confirm. What survives those qualifications is still worth having: the susceptibility is measurable in advance, the regression follows, and the interval between them is warning time.

The reframing carries a cost the article has not concealed. To build a function competent to guard the zeitgeist is to build a function competent to capture it, and the governance settlement that separates the two is demanding, fragile, and the proper subject of the next phase of work. But the discipline that has spent the better part of a century learning how to warn of strategic surprise, how to contest deception, and how to subject both to democratic oversight, is the discipline equipped to attempt it. The cognitive substrate of an open society is, in Kent's exact sense, knowledge vital to national survival. It is time it became someone's intelligence requirement.

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Notes

¹The most developed of these efforts are examined in Section 2, where each is cited in full. The point of departure here is the paradigm they share, not any single proposal; the serious strategic statement of the cognitive contest is Robert “Jake” Bebbler, *Cognitive Competition, Conflict, and War: An Ontological Approach* (Washington, DC: Hudson Institute, 2024).

²Roberta Wohlstetter, *Pearl Harbor: Warning and Decision* (Stanford, CA: Stanford University Press, 1962). Wohlstetter’s finding that the indicators of the attack were available yet irrecoverable amid the volume of competing noise is the governing analogy of this article.

³Simone Kraus [SIMKRA], “Countering Cognitive Warfare: The Need for a Defense Ontology Matrix,” *Medium*, 21 May 2025, <https://medium.com/@simone.kraus/countering-cognitive-warfare-the-need-for-a-defense-ontology-matrix-0c467488a788>. The proposal names the construct “COGNITIV3” and transposes the tactic-technique-countermeasure structure of the cyber catalogues to cognition. It is an independent online essay rather than a peer-reviewed work, and is engaged here because its framing is representative and its instinct toward an ontology is sound, not because it carries scholarly authority.

⁴The point is not that the analogy is illegitimate but that it is unproven. A technique catalogue presupposes that the unit of analysis, the discrete technique, is where the contest is decided. In cyberspace, where intrusion is event-like, that presupposition is defensible. In cognition, where the decisive effect is the slow conditioning of a population, it is precisely what is in question.

⁵Kraus, “Countering Cognitive Warfare.” The conflation of state-scale strategic subversion with the harassment of individual “targeted” persons is the diagnostic signature of a framework that lacks a level of abstraction above the technique.

⁶Bonnie Rushing, William Hersch, and Shouhuai Xu, “Cognitive Warfare: Definition, Framework, and Case Study,” arXiv preprint arXiv:2603.05222 (2026). The acute/chronic distinction and the language of re-parameterisation are the most strategically suggestive contribution of the current technical literature.

⁷Bebbler, *Cognitive Competition, Conflict, and War*. The work supplies the strategic diagnosis this article extends, though it stops short of an operational warning instrument.

⁸Vladlena Benson and John McAlaney, eds., *Cyber Influence and Cognitive Threats* (London: Academic Press, 2020); Pavel Sharikov, “AI and Information: Implications on Military Strategies,” in *AI, Information, and Global Dynamics*, ed. Fatima Roumate (Cham: Springer Nature, 2026), 99–111, doi:10.1007/978-3-032-13528-5_7. The two literatures jointly establish the supply side of the problem: injections are now cheap, abundant, and machine-produced, which is precisely why their individual interdiction cannot be the strategic answer.

⁹Richard K. Betts, “Analysis, War, and Decision: Why Intelligence Failures Are Inevitable,” *World Politics* 31, no. 1 (1978): 61–89, doi:10.2307/2009967. Betts’s thesis, that intelligence failures are failures of analysis and political receptivity rather than of collection, transfers to the cognitive domain without modification.

¹⁰Peter Gill and Mark Phythian, “What Is Intelligence Studies?” *The International Journal of Intelligence, Security, and Public Affairs* 18, no. 1 (2016): 5–19, at 5, doi:10.1080/23800992.2016.1150679. The four projects, research/historical, definitional/methodological, organisational/functional, and governance/policy, are Gill and Phythian’s; this article touches each in turn, and locates itself within the field’s standing concern with the relationship between intelligence, the state, and the broader society.

¹¹Christopher Andrew, *The Secret World: A History of Intelligence* (New Haven, CT: Yale University Press, 2018); Michael Herman with David Schaefer, *Intelligence Power in Practice* (Edinburgh: Edinburgh

University Press, 2022). Andrew’s long history and Herman’s analysis of intelligence as a form of state power jointly establish that perception management is continuous with, not external to, the intelligence function.

¹²Sherman Kent, *Strategic Intelligence for American World Policy* (Princeton, NJ: Princeton University Press, 1949); J. Ransom Clark, *Intelligence and National Security: A Reference Handbook* (Westport, CT: Praeger Security International, 2007); Loch K. Johnson, ed., *Strategic Intelligence*, 5 vols. (Westport, CT: Praeger Security International, 2007). Kent supplies the definition, Clark and Johnson the field’s mapping of strategic intelligence as a domain of practice.

¹³Hank Prunckun, *Counterintelligence Theory and Practice*, 2nd ed. (Lanham, MD: Rowman & Littlefield, 2019), 42–46. Prunckun derives a seven-proposition theory of counterintelligence and resolves its practice into four functions: deterrence and detection on the defensive side, deception and neutralisation on the offensive.

¹⁴Prunckun, *Counterintelligence Theory and Practice*, 45. The paradox of fiction names the mechanism by which a deception induces an adversary to act upon a believed fiction, which is the identical mechanism by which a cognitive-warfare campaign induces a population to act upon an engineered narrative.

¹⁵Kjetil Anders Hatlebrekke, *The Problem of Secret Intelligence* (Edinburgh: Edinburgh University Press, 2019). Hatlebrekke’s account of the epistemological problem at the heart of intelligence, the perpetual vulnerability of the knowing subject to deliberate deception, generalises directly from the service to the society.

¹⁶Wohlstetter, *Pearl Harbor*, esp. the concluding analysis of signals and noise.

¹⁷Cynthia M. Grabo, *Anticipating Surprise: Analysis for Strategic Warning* (Lanham, MD: University Press of America, 2004); Richard K. Betts, *Surprise Attack: Lessons for Defense Planning* (Washington, DC: Brookings Institution, 1982). Grabo codifies the indicator method; Betts theorises the warning-response gap, the recurrent failure of warning to convert into protective action.

¹⁸Roger Z. George, *Intelligence in the National Security Enterprise: An Introduction* (Washington, DC: Georgetown University Press, 2020), 99, on the tradecraft of alternative and competitive analysis. The principle is doubly relevant here: counter-opinion is both the analytic safeguard against a warning’s premature dismissal and, transposed to the level of the society, the very property of the substrate whose erosion the spiral of silence describes (Section 4). A polity that has lost the capacity to sustain counter-opinion has lost the social analogue of the warning function itself.

¹⁹Antonio Gramsci, *Selections from the Prison Notebooks*, ed. and trans. Quintin Hoare and Geoffrey Nowell Smith (New York: International Publishers, 1971). On the contemporary entanglement of the universities with the security and intelligence apparatus, and the corresponding vulnerability of the academy as a producer of certified knowledge, see Liam Francis Gearon, ed., *The Routledge International Handbook of Universities, Security and Intelligence Studies* (London: Routledge, 2019); and, on the historically intimate and fraught relationship between intellectuals and the intelligence state, Peter C. Grace, *The Intelligence Intellectuals: Social Scientists and the Making of the CIA* (Washington, DC: Georgetown University Press, 2026). The thinker node is not an abstraction: it is the actual institutional infrastructure of certified knowledge, and that infrastructure is itself a contested object.

²⁰Walter Lippmann, *Public Opinion* (New York: Harcourt, Brace, 1922); Edward L. Bernays, *Propaganda* (New York: Horace Liveright, 1928). Lippmann supplies the diagnosis of mediated cognition, Bernays the engineering manual.

²¹Elisabeth Noelle-Neumann, “The Spiral of Silence: A Theory of Public Opinion,” *Journal of Communication* 24, no. 2 (1974): 43–51, doi:10.1111/j.1460-2466.1974.tb00367.x. The spiral of silence is load-bearing at the society node: it explains how a small engineered shift in perceived consensus can cascade into a large shift in expressed consensus, and thence into the zeitgeist.

²²Mari-Liis Madisson and Andreas Ventsel, *Strategic Conspiracy Narratives: A Semiotic Approach*

(Abingdon: Routledge, 2021). Their analysis of the self-sealing circulation of conspiratorial frames is the clearest available account of a campaign that attacks the substrate rather than any particular belief.

²³Peter Turchin and Sergey A. Nefedov, *Secular Cycles* (Princeton, NJ: Princeton University Press, 2009); Peter Turchin, *Ages of Discord: A Structural-Demographic Analysis of American History* (Chaplin, CT: Beresta Books, 2016). The structural-demographic drivers are adopted here as warning indicators of susceptibility, not as deterministic predictors of any particular event; their forecasting validity remains a contested and open question, and the use made of them below is heuristic.

²⁴Michael Coppedge, John Gerring, Carl Henrik Knutsen, Staffan I. Lindberg, et al., “V-Dem Country-Year Dataset v16,” Varieties of Democracy (V-Dem) Project, 2026, doi:10.23696/vdemds26; Daniel Pemstein, Kyle L. Marquardt, Eitan Tzelgov, et al., “The V-Dem Measurement Model: Latent Variable Analysis for Cross-National and Cross-Temporal Expert-Coded Data,” V-Dem Working Paper, 2026. The susceptibility index below is constructed from the V-Dem Core indices; the working file is the Country-Year Full+Others release, from which the external economic covariate is also drawn.

²⁵On the analysis of intelligence beyond the Anglosphere, and on intelligence politicisation and the contest over information during political transitions in particular, see Tshepo Gwatiwa, ed., *Contemporary Intelligence in Africa* (London: Routledge, 2025). The point is methodological as well as normative: a warning instrument that holds only in consolidated Western democracies would be of limited strategic value, and the global panel is what permits the claim to be tested rather than assumed.

²⁶The operationalisation follows the logic of the autocratization literature, which dates episodes of substantial democratic regression by sustained decline in the V-Dem electoral-democracy index. The 0.10/five-year rule is varied systematically in Section 5.6 to confirm that the result is not an artefact of the threshold.

²⁷Anna Lührmann, Marcus Tannenberg, and Staffan I. Lindberg, “Regimes of the World (RoW): Opening New Avenues for the Comparative Study of Political Regimes,” *Politics and Governance* 6, no. 1 (2018): 60–77, doi:10.17645/pag.v6i1.1214. The restriction is not a tuning choice but a condition of coherence: the inflection is defined as the erosion of a democratic-epistemic order, and only polities that possess such an order are at risk of the event. Closed and electoral autocracies sit at or near the floor of the electoral-democracy scale and cannot register a decline of the required magnitude, the familiar room-to-fall problem in the study of regime change.

²⁸Dan Hansén, “Navigating Incompatible Imperatives in Intelligence Affairs: Bureaucracy, Flexibility and Pragmatism,” *Intelligence and National Security* 41, no. 1 (2026): 158–181, doi:10.1080/02684527.2025.2595032. The design of a cognitive-warning function is precisely an exercise in navigating such incompatible imperatives, and Hansén’s analysis cautions against the assumption that an optimal institutional form exists to be specified rather than continuously negotiated.

²⁹Joshua Rovner, *Fixing the Facts: National Security and the Politics of Intelligence* (Ithaca, NY: Cornell University Press, 2011). Rovner’s anatomy of politicisation is the indispensable cautionary framework for any proposal to institutionalise a cognitive-warning function, whose product, an assessment of the health of public reasoning, is more exposed to political capture than almost any other intelligence product.

³⁰Christopher Moran, *Classified: Secrecy and the State in Modern Britain* (Cambridge: Cambridge University Press, 2013). Moran’s account of the British state’s transition from the suppression of information to its active management is the essential historical precedent for any contemporary proposal that the state should concern itself with the condition of public knowledge, and a standing warning about where such concern tends to lead.

³¹Christopher Moran, *Company Confessions: Secrets, Memoirs, and the CIA* (New York: St. Martin’s Press, 2016). The history of the contest over the disclosure of the CIA’s record establishes both that the security state will seek to manage public knowledge and that this management is itself contestable through

the institutions of a free press and a porous bureaucracy, which is the very dynamic any cognitive-warning function must be designed to preserve rather than to suppress.

³²Christopher Andrew, Richard J. Aldrich, and Wesley K. Wark, eds., *Secret Intelligence: A Reader*, 2nd ed. (London: Routledge, 2020). The Reader's organisation of the field around the intelligence cycle, counter-terrorism and security, ethics and accountability, and intelligence in the new warfare maps the terrain on which a cognitive-warning function would have to be located and governed; George, *Intelligence in the National Security Enterprise*, supplies the institutional grammar of how such a function would sit within the wider enterprise.

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Appendix: Methods, Variables, and Replication

This appendix documents the construction of the susceptibility index, the exact V-Dem variables used, the full list of inflection onsets, and the robustness and mechanism analyses summarised in Section 5. It is intended also to serve as an online supplement; the analysis is fully reproducible from the scripts named in A.7.

A.1 Variables

All six index components are V-Dem version 16 Country-Year Core indices on a zero-to-one scale. Five are oriented so that higher values denote a healthier substrate; regime corruption is oriented so that higher values denote a more corrupt regime and is therefore entered as its complement. Coverage in the analysis window, democracies from 1970 to 2019, is effectively complete: each of the six components is present for 100 per cent of country-years and GDP per capita for 99.9 per cent, so the five-of-six availability rule below rarely binds in the estimation sample and matters chiefly on the wider panel.

Table 8. Variables used in the analysis (V-Dem v16; codebook labels).

Code	Label	Scale / coding	Role
v2x_polyarchy	Electoral democracy index	0–1, higher more democratic	Outcome
v2xme_altinf	Alternative sources of information index	0–1, higher more plural	Informational (<i>F</i>)
v2x_freexp_altinf	Freedom of expression and alternative sources of information index	0–1, higher freer	Informational (<i>F</i>)
v2x_rule	Rule of law index	0–1, higher stronger	Institutional (<i>L</i>)
v2xlg_legcon	Legislative constraints on the executive	0–1, higher more constraint	Institutional (<i>L</i>)
v2x_jucon	Judicial constraints on the executive	0–1, higher more constraint	Institutional (<i>L</i>)
v2xnp_regcorr	Regime corruption index	0–1, higher more corrupt; entered as 1– value	Institutional (<i>L</i>)
v2x_regime	Regimes of the World	0 closed aut.; 1 electoral aut.; 2 electoral dem.; 3 liberal dem.	Population (≥ 2)
e_gdppc	GDP per capita (latent estimate)	logged, lagged one year	Control
e_regiongeo	Region (geographic)	categorical	Robustness FE

A.2 Index construction

Let the six healthy-substrate components of a country-year be $c_1, \dots, c_6$, with regime corruption entered as $1 - \text{v2xnp_regcorr}$. The institutional-epistemic susceptibility index is

$$\text{IESI} = 1 - \frac{1}{|S|} \sum_{i \in S} c_i,$$

where S is the set of components present in that country-year, computed only when $|S| \geq 5$. The components are equally weighted and are *not* re-standardised before averaging, because they already share a common zero-to-one scale; only the resulting index is standardised to zero mean and unit variance within the estimation sample when it enters the model, so that its coefficient reads as the effect of a one-standard-deviation increase. The accountability-only variant uses the four institutional components alone, none of which enters the electoral-democracy outcome. The informational sub-index used in the mechanism test (A.6) uses the two informational components alone, and the institutional sub-index the four institutional components.

A.3 Estimator, population, and outcome

The estimator is a discrete-time event-history (survival) logit. The population is the world's democracies, defined by Regimes of the World ≥ 2 . The dependent event is an inflection onset, coded where the electoral-democracy index declines by at least 0.10 over the subsequent five years and the prior year was not already within such a decline; the onset is dated to the first year of the episode and years internal to a decline are removed from the risk set. Standard errors are clustered by country. The out-of-sample test estimates on country-years through 2009 and predicts 2010–2019. Full specification is given in Sections 5.3 and 5.4.

A.4 Inflection onsets

Table 9 lists all 60 onsets entering the estimation sample.

A.5 Robustness

Each row below re-estimates the main in-sample specification (full IESI, democracies, 1970–2019) under a perturbation; the reported quantity is the odds factor for a one-standard-deviation increase in susceptibility.

- **Leave-one-component-out.** Dropping each of the six components in turn moves the odds factor only within 1.70 to 1.92, all significant at $p < 0.002$ (`altinf` 1.80, `freexp` 1.77, `rule` 1.75, `legcon` 1.92, `jucon` 1.74, `clean-government` 1.71). No single component, and in particular no single informational component, drives the result.
- **Fixed effects.** Decade fixed effects leave the estimate unchanged (odds factor 1.80, $p < 0.001$); region-and-decade fixed effects attenuate it modestly but it remains significant (1.66, $p = 0.012$). Region-only fixed effects did not converge, a rare-event separation across many small region cells rather than a substantive null.
- **Alternative lags.** Using the susceptibility index at $t - 1$, $t - 2$, and $t - 3$ yields odds factors

Table 9. Inflection onsets, democracies ($\text{RoW} \geq 2$), 1970–2019: country and onset year ($N = 60$).

Country	Yr	Country	Yr	Country	Yr
Armenia	2019	Hungary	2008	Peru	1987
Belarus	1992	India	2010	Peru	2018
Benin	2014	Ivory Coast	2019	Philippines	2000
Bolivia	2014	Lesotho	1993	Philippines	2015
Botswana	2015	Libya	2013	Poland	2011
Botswana	2017	Maldives	2009	Romania	2019
Brazil	2013	Mali	2007	Serbia	2008
Burkina Faso	2010	Mali	2015	Slovenia	2015
Burkina Faso	2017	Mauritius	2015	Solomon Is.	1995
Croatia	2012	Mexico	2018	South Korea	2018
Ecuador	2005	Moldova	2000	Thailand	2001
El Salvador	2015	Moldova	2011	Tunisia	2016
Fiji	1982	Mongolia	2018	Türkiye	1975
Fiji	1995	Nicaragua	1995	Türkiye	2004
Georgia	2019	Nicaragua	2002	Türkiye	2009
Greece	2016	Niger	1993	Ukraine	1994
Guatemala	2017	Niger	2004	Ukraine	2007
Guinea-Bissau	2019	Niger	2014	Ukraine	2019
Guyana	2016	Niger	2018	Venezuela	1995
Honduras	2004	N. Macedonia	2006	Zambia	2011

of 1.80, 1.89, and 1.77 respectively, all $p < 0.001$; the signal is present across the plausible warning horizon.

- **Out-of-sample diagnostics.** On the 2010–2019 test window (853 country-years, base rate 0.040): ROC-AUC 0.70; precision-recall AUC 0.091 against a no-skill baseline of 0.040; Brier score 0.038 against a climatological 0.038. Calibration by predicted-risk quintile is reported in Table 5.
- **Onset-definition grid.** Varying the decline threshold (0.08/0.10/0.12) and window (4/5/6 years) leaves the odds factor in 1.70–1.88, all $p < 0.001$ (Table 6).

A.6 Mechanism test

The theory holds that the informational substrate degrades before the institutional order. Two tests examine this, and both decline to confirm it. A cross-lagged panel model with country fixed effects and country-clustered standard errors regresses each susceptibility dimension on the prior-year value of both: the lead of informational on institutional susceptibility is not significant ($\beta = -0.009$, $p = 0.45$), and neither is the reverse ($\beta = +0.039$, $p = 0.33$). A predictive horse-race that enters the informational and institutional sub-indices jointly in the

onset logit assigns the cleaner weight to the institutional dimension (odds factor 1.55, $p = 0.05$) over the informational (odds factor 1.21, $p = 0.20$). The data therefore do not establish an informational-leads-institutional ordering; the substrate-first claim is retained as a theoretical conjecture and is reported as such in Section 5.7. This is a candid null, and it bounds the empirical contribution of the article rather than extending it.

A.7 Replication

Two scripts reproduce every number above. `analysis/vdem_backtest.py` builds the panel and produces the main models, the onset-definition grid, and the trajectory figure; `analysis/vdem_appendix.py` produces the onset list, the out-of-sample diagnostics, the leave-one-component-out and fixed-effects robustness, the alternative-lag estimates, and the mechanism test. The data are the V-Dem v16 Country-Year Core and Full+Others releases, publicly available from the V-Dem Institute ([doi:10.23696/vdemds26](https://doi.org/10.23696/vdemds26)). Computation is in Python with `pandas` and `statsmodels`; standard errors are cluster-robust by country throughout.